EVS – Unit III

Important 10-Mark Questions with Easy Answers

Prepared from your uploaded notes. These two broad questions cover several major topics from Unit III. Study the headings and bullet points for exam revision.

Question 1: Explain the major types of environmental pollution, their sources and control measures.

Introduction

Environmental pollution is an undesirable change in the physical, chemical or biological characteristics of air, water and land. It harms living organisms and ecosystems. The major types are air, water, soil/land, noise, thermal, radioactive and marine pollution.

1. Air Pollution

- Sources: Industries, vehicles, burning coal and wood, agricultural activities, forest fires and dust storms.
- Control: Use solar and wind energy, public transport and electric vehicles; install filters in industries; avoid open burning; plant green belts; monitor emissions.

2. Water Pollution

- Sources: Domestic sewage, industrial wastewater, fertilizers, pesticides, oil spills, solid waste, mining and soil erosion.
- Effects: Reduced dissolved oxygen, excessive algal growth, death of aquatic organisms, water-borne diseases and bioaccumulation.
- Control: Treat sewage, establish Effluent Treatment Plants (ETPs), reduce chemical use, prevent waste dumping, conserve water and reuse treated wastewater.

3. Soil/Land Pollution

- Sources: Excessive fertilizers and pesticides, industrial and hazardous waste, plastics, e-waste, mining and improper municipal waste disposal.
- Control: Encourage organic farming, reduce chemical use, segregate and recycle waste, compost organic waste, treat hazardous waste and restore polluted land.

4. Noise Pollution

- Sources: Road, rail and air traffic, industrial machines, construction, loudspeakers and generators.
- Control: Use silencers and sound-proof enclosures, maintain machines, limit horns and loudspeakers, create silent zones and enforce noise standards.

5. Thermal Pollution

- Sources: Hot water from thermal and nuclear power plants, industrial cooling and removal of vegetation near water bodies.
- Control: Use cooling towers and ponds, recycle cooling water, cool wastewater before discharge and maintain vegetation near water bodies.

6. Radioactive Pollution

- Sources: Nuclear plants and accidents, unsafe radioactive waste disposal, medical/research uses and radioactive-ore mining.
- Control: Store and dispose of radioactive waste safely, shield radioactive materials, monitor nuclear facilities and follow radiation-safety rules.

7. Marine Pollution

- Sources: Oil spills, plastics and solid waste, sewage, industrial wastewater, agricultural runoff and fishing activities.
- Control: Reduce plastic use, treat wastewater, prevent oil spills, promote sustainable fishing and monitor marine ecosystems.

General Pollution-Control Strategies

- Follow the 3Rs: Reduce, Reuse and Recycle.
- Use cleaner technologies and renewable energy.
- Improve public transport and sustainable agriculture.
- Enforce environmental laws and conduct environmental impact assessments.
- Increase environmental awareness and community participation.
- Follow the principle: “Polluter Pays.”

Conclusion

Pollution can be reduced by preventing pollutants at the source, treating waste safely, using cleaner technologies, following environmental laws and involving the public.

Question 2: Explain climate change, global warming, the greenhouse effect, ozone depletion and acid rain.

1. Climate Change

Climate change means long-term changes in global temperature, rainfall and weather patterns. Human activities are a major cause.

Causes

- Burning coal, petroleum and natural gas.
- Deforestation and land-use changes.
- Industrial activities, transport, agriculture and livestock.
- Increasing greenhouse-gas emissions.

Effects

- Increase in global temperature.
- Melting of glaciers and polar ice; rising sea levels.
- Changes in rainfall patterns.
- More heatwaves, floods, droughts and cyclones.
- Loss of biodiversity and threats to agriculture, water resources and human health.

Control / Response

- Reduce emissions and use renewable energy.
- Protect forests and reduce deforestation.
- Improve energy efficiency and use public transport.
- Take mitigation measures to reduce causes and adaptation measures to manage unavoidable impacts.

2. Global Warming

Global warming is the long-term rise in the average temperature of the Earth's surface and atmosphere, mainly due to increased greenhouse gases.

- Causes: Fossil-fuel burning, deforestation, industry, agriculture and transportation.
- Effects: Higher temperatures, melting ice, sea-level rise, changing rainfall, heatwaves and pressure on food and water resources.

3. Greenhouse Effect

The greenhouse effect is a natural process in which certain gases absorb and re-emit heat, keeping Earth warm enough for life. Excess greenhouse gases strengthen this effect and cause additional warming.

Major greenhouse gases

- Carbon dioxide (CO■), methane (CH■), nitrous oxide (N■O), water vapour, ozone (O■) and fluorinated gases.

Key point: The natural greenhouse effect supports life, but the enhanced greenhouse effect contributes to current global warming.

4. Ozone Depletion

Ozone depletion is the reduction of ozone in the stratosphere. The ozone layer absorbs much of the Sun's harmful ultraviolet (UV) radiation.

Causes

- Chlorofluorocarbons (CFCs), halons, carbon tetrachloride, methyl chloroform and other ozone-depleting substances.

Effects

- Greater exposure to harmful UV-B radiation.
- Higher risk of skin cancer and eye damage, including cataracts.
- Suppression of immune functions; damage to crops, plants and marine organisms.

Control

- Follow the Montreal Protocol, use safer alternatives and manage refrigerants properly.

5. Acid Rain

Acid rain is precipitation that becomes more acidic than normal due to pollutants, mainly sulphur dioxide (SO■) and nitrogen oxides (NO■). These react in the atmosphere to form acids that reach Earth through rain, snow, fog or dry deposition.

Sources / Causes

- Coal-fired power plants, industries, burning petroleum, motor vehicles and metal-processing industries.

Effects

- Changes soil chemistry and reduces essential nutrients.
- Lowers the pH of lakes and streams, harming aquatic organisms.
- Damages leaves and plant growth.
- Speeds up damage to limestone, marble and buildings.

Control

- Use low-sulphur fuels and renewable energy.
- Install flue-gas desulfurization systems.
- Reduce vehicle emissions, improve energy efficiency and enforce industrial emission standards.

Conclusion

Climate change and related environmental problems can be reduced by lowering greenhouse-gas and other pollutant emissions, protecting forests, using clean energy and following environmental regulations.

Quick exam tip: For a 10-mark answer, write a short definition/introduction, use clear headings, explain causes or sources, list effects where relevant, give control measures, and finish with a 2–3 line conclusion.